

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. The exact information state is the resolved history together with the controlled posterior law of the complete microstate. Each transition consists of normalized controlled prediction followed by disintegration with respect to the newly observed block, yielding measurable strategy laws and an exact posterior-state DPP. Conditional source estimates prove the adaptive kinetic Isaacs limit. Typical posterior density odds have a directed relative-entropy rate, whereas expected posterior error has the Chernoff testing rate. The canonical experiment is centred at the exact finite-volume mean and satisfies LAN and a finite-dimensional Bernstein–von Mises theorem.

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1. ROUND-SIX POSITIVE CLOSURE: OBSERVED POSTERIOR GAMES AND CANONICAL STATISTICAL TANGENTS

The exact finite information state is the complete resolved history together with the posterior law of the microscopic hard-sphere state under the already controlled law. Every posterior transition contains two distinct steps:

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controlled prediction and disintegration by the newly observed block. The expected posterior-error exponent is Chernoff/testing information, whereas the typical log posterior odds have the directed relative-entropy rate. Local asymptotic normality is centred at the exact finite-volume mean.

1.1. Resolved observations and posterior kernels. Let Z_t^ε be the complete hard-sphere microstate and let Y_k^ε be the graph-completed resolved observation on the block $[t_k, t_{k+1}]$. The observation may contain finitely many density and actual-contact cylinders and is a Borel map of the block path. Put

$$\mathcal{F}_{t_k}^{\text{res}} = \sigma(Y_0^\varepsilon, \dots, Y_{k-1}^\varepsilon), \quad \Pi_k^\varepsilon = \text{Law}(Z_{t_k}^\varepsilon \mid \mathcal{F}_{t_k}^{\text{res}}).$$

Regular conditional laws exist because both microstate and graph-completed observation spaces are standard Borel.

For compact controls $u \in U$, $v \in V$, let $A_k^{u,v}(Z_{[t_k, t_{k+1}]})$ be a bounded particle/contact work. Given a posterior π , define the exact conditional log normalizer

$$\Lambda_{k,\varepsilon}^{u,v}(\pi) = \frac{1}{\mu_\varepsilon} \log \int e^{\mu_\varepsilon A_k^{u,v}(Z)} \mathbf{P}_{\Delta_k}^\varepsilon(\pi, dZ),$$

where $\mathbf{P}_{\Delta_k}^\varepsilon(\pi, \cdot)$ is the deterministic block-path law obtained by starting the Liouville flow from π . The controlled predictive block law is

$$\bar{\mathbf{P}}_k^{u,v}(\pi, dZ) = e^{\mu_\varepsilon (A_k^{u,v}(Z) - \Lambda_{k,\varepsilon}^{u,v}(\pi))} \mathbf{P}_{\Delta_k}^\varepsilon(\pi, dZ).$$

Let $O_k(Z) = Y_k$ be the new observation. Disintegrate $\bar{\mathbf{P}}_k^{u,v}$ with respect to O_k and denote the conditional terminal microstate law by

$$\mathcal{B}_{k,\varepsilon}^{u,v}(\pi, y; dZ').$$

Lemma 1.1 (Controlled prediction–observation Bayes kernel). *For every posterior π , the predictive density above has total mass one. The map*

$$(\pi, u, v) \longmapsto \bar{\mathbf{P}}_k^{u,v}(\pi, \cdot)$$

is Borel, and a jointly Borel version of $\mathcal{B}_{k,\varepsilon}^{u,v}$ exists. Under the controlled law the posterior update is exactly

$$\Pi_{k+1}^{\varepsilon,u,v} = \mathcal{B}_{k,\varepsilon}^{u,v}(\Pi_k^{\varepsilon,u,v}, Y_k^\varepsilon; \cdot).$$

Thus the update includes conditioning on the new observation; it is not merely the tilted predictive pushforward.

Proof. The definition of $\Lambda_{k,\varepsilon}^{u,v}$ normalizes the predictive law. Boundedness and Borel dependence of the work, together with continuity of deterministic pushforward on bounded cylinders, give Borel dependence of the predictive kernel. Standard Borel disintegration provides a jointly measurable posterior kernel after modifying one null set for a countable generating algebra. Applying Bayes' formula first to the exponential block change of law and then to the sigma-field generated by O_k gives the stated update. $\square$

1.2. Strategy-dependent laws and the exact DPP. A relaxed control is a probability on U or V . An Elliott–Kalton strategy maps the opponent’s past relaxed controls and the resolved history to the next relaxed control, nonanticipatively. The information state is

$$S_k^\varepsilon = (H_k^\varepsilon, \Pi_k^\varepsilon),$$

where $H_k^\varepsilon = (Y_0^\varepsilon, \dots, Y_{k-1}^\varepsilon)$.

Theorem 1.2 (Canonical strategy-law construction). *For every pair of nonanticipative relaxed strategies and every initial posterior, the Bayes kernels of Lemma 1.1 determine a unique controlled probability law on histories, posteriors, controls, and microstates. Under that law each block likelihood increment is conditionally mean one with respect to the already controlled posterior. The family is stable under concatenation at stopping grid times.*

Proof. At the first block, evaluate the strategies at the initial history and use the normalized predictive/observation kernel. Inductively, the realized history and posterior determine the next relaxed controls and hence the next kernel. Ionescu–Tulcea gives a unique path law. The normalizer is evaluated under the posterior generated by all preceding controlled kernels and observations, so the next likelihood increment has conditional mean one. Disintegration at a stopping grid time and restarting the same recursion proves concatenation. $\square$

Let the terminal payoff be bounded and let the running reward be continuous in the information state and relaxed controls.

Theorem 1.3 (Exact posterior-state dynamic programming). *The upper and lower finite-grid game values are Borel functions of (H_k, Π_k) and satisfy the exact dynamic programming recursion on that state. Compactness of relaxed control spaces gives measurable selectors. A time-zero preparation action, a reward-only feedback action, and an adaptive law action define three different games:*

- (i) *a preparation action is chosen once and remains a static state coordinate through all blocks;*
- (ii) *reward-only feedback changes the running payoff but not the transition kernel;*
- (iii) *adaptive law control changes the normalized predictive kernel and hence the posterior transition.*

No pointwise Isaacs equation is assigned to the first game unless the static preparation action is retained in the state.

Proof. Condition the value at the next grid time. The state (H_{k+1}, Π_{k+1}) is a measurable function of the current state, relaxed controls, and new observation through Lemma 1.1. The tower property under the unique law of Theorem 1.2 gives the recursion. Probability measures on compact action sets are compact, and continuity of bounded work/reward makes the one-step

value lower/upper semicontinuous, so measurable minimax selectors exist. The three timing statements follow directly from which coordinate of the kernel or payoff the action changes. $\square$

1.3. Posterior trace classes and the kinetic contraction. Let

$$\|\pi\|_{\text{tr}} = \|G(\pi)\|_{\alpha,\beta}^{\text{tr}}$$

be the B2 trace-hierarchy norm. Conditional expectation gives

$$\mathbb{E}\|\Pi_k^\varepsilon\|_{\text{tr}} \leq C_k \|\Pi_0^\varepsilon\|_{\text{tr}} e^{Ck\Delta e^{CR}}$$

for bounded control sources. Thus posterior trace norms need not be uniformly bounded for every rare observation, but they have exponential/entropy containment under every strategy.

Lemma 1.4 (Strategy-uniform conditional source charts). *On each posterior trace ball, the next-block conditional particle/contact pressure is the B2 pressure with that posterior hierarchy as initial state and has the same analytic radius and derivative bounds, with constants depending only on the ball. Under bounded strategies, posterior paths leave an increasing sequence of such balls with exponentially vanishing probability.*

Proof. The B2 trace-class theorem is linear in the initial complete hierarchy and its operator bounds depend only on its trace norm. Conditioning changes the initial hierarchy but not the local hard-sphere dynamics. Hence the same connected expansion applies on a trace ball. The normalized block likelihood and Bayes disintegration preserve the average trace norm; the entropy inequality applied to its exponential Lyapunov truncations gives exponential containment uniformly over bounded control works. Iteration over finitely many blocks gives the claim. $\square$

Let $q^{u,v}$ be the limiting positive bounded actual-contact multiplier selected by the control. Its kinetic Hamiltonian is

$$\mathbb{H}^{u,v}(f, p) = \langle f, v \cdot \nabla p \rangle + \int q^{u,v} (e^{\Delta p} - 1) dA_f + c(f, u, v).$$

Theorem 1.5 (Posterior-to-kinetic contraction). *On every finite control grid, conditionally on the resolved density/contact history, the posterior finite marginals satisfy a uniform conditional Laplace principle whose unique regular minimizer is the controlled B2 biased hierarchy. Consequently the posterior coordinate contracts, in finite correlation projections, to that hierarchy and its one-particle density. As the grid mesh tends to zero, the game values converge to the unique B4 viscosity solution of*

$$\partial_t U + \sup_u \inf_v \mathbb{H}^{u,v} \left(f, \frac{\delta U}{\delta f} \right) = 0$$

when the Isaacs equality holds. The exact finite information state remains (H, Π) ; density reduction is a proved kinetic limit, not a finite-scale identity.

Proof. On one block, apply the B2 normalized bounded-source tilt starting from the current posterior trace hierarchy. The conditional pressure derivatives and strict quotient covariance identify one regular biased hierarchy, and the conditional large-deviation upper and lower bounds give exponential concentration there. Lemma 1.4 makes the estimates uniform on exponentially containing trace balls. Induct over the finite grid, carrying the posterior update with the new observations. Finite correlation projections then converge to the controlled hierarchy.

The discrete DPP is consistent with the displayed Hamiltonian by the B2 one-block pressure. B4 compact containment and comparison identify the upper and lower half-relaxed limits as the mesh vanishes. Isaacs equality identifies the two game values. $\square$

1.4. Typical odds and expected posterior error. Let $\{\mathbb{P}_{\theta,\varepsilon} : \theta \in \Theta \subset \mathbb{R}^d\}$ be a compact canonical exponential path family with

$$\frac{d\mathbb{P}_{\theta,\varepsilon}}{d\mathbb{P}_{0,\varepsilon}} = \exp\{\mu_\varepsilon(\theta \cdot X_\varepsilon - Q_\varepsilon(\theta))\}.$$

Assume the prior has a continuous density bounded above and below near the true interior parameter θ_0 . Posterior densities are always taken with respect to Lebesgue measure or this prior; no ratio to the mass of a singleton is used.

For $\theta \neq \theta_0$, define the Chernoff information

$$\mathcal{C}(\theta_0, \theta) = - \inf_{0 \leq s \leq 1} [Q((1-s)\theta_0 + s\theta) - (1-s)Q(\theta_0) - sQ(\theta)].$$

Theorem 1.6 (KL odds and Chernoff posterior-error exponents). *Under $\mathbb{P}_{\theta_0,\varepsilon}$, the typical posterior log-density ratio obeys*

$$\frac{1}{\mu_\varepsilon} \log \frac{r_T^\varepsilon(\theta)}{r_T^\varepsilon(\theta_0)} \longrightarrow -\mathcal{K}(\theta_0 \parallel \theta)$$

in probability, locally uniformly away from nonidentifiable parameters. For every neighborhood O of θ_0 ,

$$\limsup_{\varepsilon \rightarrow 0} \frac{1}{\mu_\varepsilon} \log \mathbb{E}_{\theta_0} \int_{O^c} r_T^\varepsilon(\theta) d\theta \leq - \inf_{\theta \in O^c} \mathcal{C}(\theta_0, \theta).$$

The right side is strictly negative when the family is identifiable.

Proof. The log likelihood ratio divided by μ_ε converges by the pressure law of large numbers to its expectation, which is minus the directed relative-entropy rate. Prior-density ratios are subexponential, giving the first assertion.

For expected posterior error, use $x/(1+x) \leq x^s$, $0 \leq s \leq 1$, for each likelihood ratio and optimize in s . The exact exponential-family moment is the Chernoff pressure displayed above. Cover the compact complement of O by finitely many neighborhoods on which pressure and prior density vary by at most $o(1)$; analytic compactness makes the cover independent of ε . Integrating the pointwise bounds and optimizing proves the exponent. Strict convexity on the identifiable quotient makes the infimum positive. $\square$

1.5. Exact-centred LAN and Bernstein–von Mises. Define

$$\Delta_\varepsilon = \sqrt{\mu_\varepsilon}(X_\varepsilon - DQ_\varepsilon(\theta_0)), \quad I_\varepsilon = D^2Q_\varepsilon(\theta_0).$$

Theorem 1.7 (Finite-volume-centred canonical LAN). *Uniformly for h in compact subsets of $\mathbb{R}^d$,*

$$\log \frac{d\mathbb{P}_{\theta_0+h/\sqrt{\mu_\varepsilon},\varepsilon}}{d\mathbb{P}_{\theta_0,\varepsilon}} = h \cdot \Delta_\varepsilon - \frac{1}{2} h^T I_\varepsilon h + o_{\mathbb{P}}(1).$$

The score is exactly centred for every finite ε . Moreover $\Delta_\varepsilon \Rightarrow N(0, I)$, where $I = D^2Q(\theta_0) > 0$ on the identifiable quotient.

Proof. Taylor-expand the exact finite pressure at $\theta_0 + h/\sqrt{\mu_\varepsilon}$. The linear term is $DQ_\varepsilon(\theta_0)h/\sqrt{\mu_\varepsilon}$ and cancels exactly with the chosen score centre. Uniform third Cauchy bounds on a larger complex chart make the remainder $O(|h|^3/\sqrt{\mu_\varepsilon})$. The B3 exact-centred Gaussian theorem gives the score limit and pressure derivative convergence gives $I_\varepsilon \rightarrow I$. $\square$

Theorem 1.8 (Canonical Bernstein–von Mises theorem). *Assume the canonical chart is finite dimensional, identifiable, the prior density is positive and continuous at θ_0 , and the compact complements admit the Chernoff tests of Theorem 1.6. Then the posterior law of $h = \sqrt{\mu_\varepsilon}(\theta - \theta_0)$, after centring by $I^{-1}\Delta_\varepsilon$, converges in total variation to $N(0, I^{-1})$.*

Proof. The Chernoff bound makes posterior mass outside every fixed neighborhood exponentially small. On balls $|h| \leq M_\varepsilon$ with $M_\varepsilon \uparrow \infty$ and $M_\varepsilon^3/\sqrt{\mu_\varepsilon} \rightarrow 0$, the LAN remainder is uniform. The prior ratio converges uniformly to one. Completing the square in the LAN likelihood gives the Gaussian density with centre $I^{-1}\Delta_\varepsilon$; dominated convergence gives total variation on the growing ball. Positive definiteness gives Gaussian tail control, and the Chernoff tests control the complement. $\square$

1.6. Smooth and discrete saddle envelopes. On a smooth finite-dimensional preparation manifold, strict signed curvature of the preparation cost plus the B3 quotient covariance gives an invertible saddle Hessian. The envelope Hessian is the corresponding Schur complement. For finite action sets the value is an active-branch directional envelope and no smooth Hessian is asserted at branch crossings.

Theorem 1.9 (Round-six C1 closure). *The finite hard-sphere control problem has an exact posterior-history information state, normalized strategy-dependent laws, and an observed Bayes DPP. Its adaptive kinetic limit is the Isaacs semigroup obtained from the B2 controlled actual-contact pressure and B4 comparison. Typical posterior odds have the KL rate, expected posterior error has the Chernoff rate, and the canonical statistical experiment satisfies exact-centred LAN and the stated Bernstein–von Mises theorem.*

Proof. The exact game follows from Theorems 1.2 and 1.3; its kinetic limit is Theorem 1.5. The statistical statements are Theorems 1.6, 1.7, and 1.8.

The timing distinctions and saddle calculus were proved on their respective exact state spaces above. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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